QUESTION 1

Why is bit manipulation critical in embedded systems?

Answer is Wrong **Score: 0 / 1**

Options

Wrong

A. To optimize floating-point calculations

Expected

B. Because hardware control often depends on individual bits

C. To replace memory allocation

D. To simplify pointer usage

FEEDBACK

Solution: Embedded systems control hardware where a single bit can act as a switch. Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=49

QUESTION 2

How is an individual bit set within a register?

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Not Attempted **Score: 0 / 1**

Options

Expected

A. Using OR operation with a bit mask

B. Using AND operation with a bit mask

C. Using NOT operation on the whole word

D. Using shifting alone

FEEDBACK

Solution: To set a bit, the data is ORed with a bitmask having a 1 at that position. Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=303

QUESTION 3

What method is used to clear a specific bit?

Not Attempted **Score: 0 / 1**

Options

A. XOR with the bit mask

Expected

B. AND with an inverted bit mask

C. OR with the bit mask

D. NOT on the word

FEEDBACK

Solution: Clearing uses AND with the complement of the bitmask, forcing that bit to 0.

Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=497

QUESTION 4

Which logical operation toggles a bit's value?

Not Attempted **Score: 0 / 1**

Options

A. OR

B. AND

Expected

C. XOR

D. Shift left

FEEDBACK

Solution: XOR flips the target bit—0 becomes 1 and 1 becomes 0. Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=634

QUESTION 5

How can we check if a particular bit is set?

Not Attempted **Score: 0 / 1**

Options

Expected

A. AND data with a bitmask and check result

B. OR data with a bitmask and check result

C. XOR data with a bitmask and check result

D. Shift data left and test

FEEDBACK

Solution: Using AND with a bitmask isolates the bit to test if it's set. Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=741

QUESTION 6

What does volatile ensure for memory-mapped I/O?

Not Attempted **Score: 0 / 1**

Options

A. Prevents race conditions

Expected

B. Forces the compiler to always access the actual hardware

C. Automatically clears registers

D. Disables interrupts

FEEDBACK

Solution: volatile stops caching/optimization, ensuring every read or write goes to the real hardware Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=869

QUESTION 7

Why are atomic operations important when modifying bits?

Not Attempted **Score: 0 / 1**

Options

A. To speed up pointer arithmetic

Expected

B. To prevent race conditions during read-modify-write

C. To avoid memory leaks

D. To enable wider addresses

FEEDBACK

Solution: Atomic operations ensure no other process changes data between read and write, avoiding race conditions.

Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=951

QUESTION 8

What does bit-banding achieve in ARM Cortex processors?

Not Attempted **Score: 0 / 1**

Options

Expected
A. Assigns unique addresses to individual bits

Expected
B. Simplifies setting/clearing bits

C. Expands memory size

Expected
D. Avoids creating bit masks

FEEDBACK

Solution: Bit-banding gives each bit a unique address, making direct atomic bit manipulation easier without masks.

Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=1102

QUESTION 9

What is a pointer in C?

Not Attempted **Score: 0 / 1**

Options

Expected

A. A variable that stores another variable's address

B. A function to allocate memory

C. A type of register

D. A memory-mapped peripheral

FEEDBACK

Solution: A pointer holds the address of another variable and allows indirect access. Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=1334

QUESTION 10

Which are best practices when using pointers?

Not Attempted **Score: 0 / 1**

Options

Expected
- [ ] A. Always initialize pointers

Expected
- [ ] B. Check pointers against NULL before use

Expected
- [ ] C. Free allocated memory

- [ ] D. Avoid using pointers in arrays

FEEDBACK

Solution: The lecture highlights initializing, validating, and freeing pointers as essential best practices

Link: https://youtu.be/WhGOuV5VsV4?feature=shared&t=2128